Relational model

What is a DB model?
- format data is in
- how data relates to itself
- how it is manipulated

1969/1970 Edgar Codd
- he wrote papers introducing relnl model

Before Codd (BC):
2 models were dominant
- network model - hardcoded relnships wherever they happened

name	address	class
Dave	here	
Amy	there	

class no
334
111

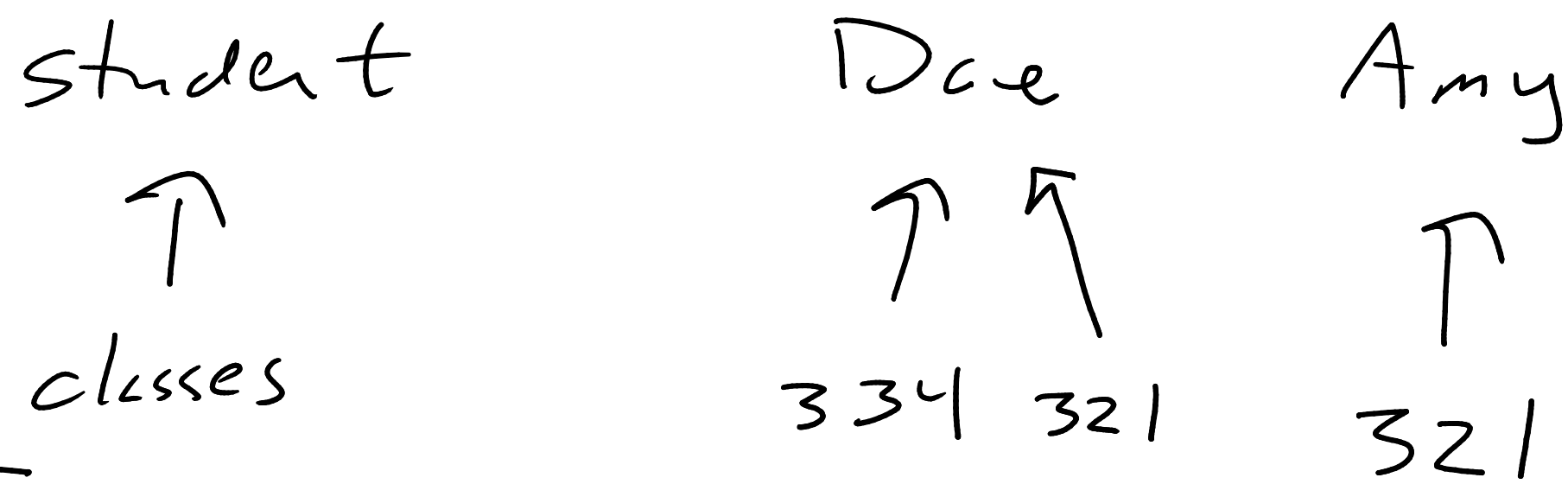
favorite food
pizza
gyros

To write queries, you wrote lots of code

You had inconsistencies, you had duplicated code

Masses were crying out for a query language

- hierarchical model - everything was in subcategories



Codd: "There's a better way."

All your relationships are data.

name	address	class
Dave	here	cs334-01

class no	name
cs334-01	DB sys

Acolytes who worked to build s/w
to implement vision 1970s

① Berkeley
Team had a grant to do graphics work
"Interactive Graphics Retrieval Systems"
→ INGRES ← DB system
- created a query language: QUEL

② IBM research
→ built System R
→ created a query language: SEQUEL
↙ changed
SQL

Time passed...
System R → IBM dB2 (commercial)

INGRES finally grudgingly added SQL to their system

→ renamed PostgreSQL

Rehl is not the only model

In the 1990s, people got upset because they weren't getting performance they wanted

1. the existing rehl systems didn't care about parallelisms
2. belief that rehl was the problem
3. people didn't like SQL much.

Movement: "no SQL" databases

↳ actually non-rehl DB systems

In the 2010s - the rehl s/w caught up (mostly)

Reln database: collection of relns

Reln: table (2-d) - rows/cols

- more mathematically:

set of tuples

Relation schema - reln design
- what colums exist

Relation instance - actual example of data

(Q about does order matter?)

In theory defn of reln -

set of tuples

Keys: purpose is to be able to distinguish rows

A set of attributes (cols) is a <u>candidate key</u> for a reln if

(1) no two distinct tuples are allowed to have the same values for those attrs

(2) not true for a subset of those attrs.

You can't determine this by looking at a reln instance.

<u>It's</u> baked into the schema.

<u>Primary key</u> - one candidate key in particular designated as primary

(rel systems make it easy to do a primary key, and easy to create other candidate keys, but terminology is inconsistent)

<u>Foreign key</u> - set of attrs in one reln that is used to refer to a primary key in another reln

In our recipe example...

rid is a foreign key in Assignments, referencing Recipes

Query languages

Codd invented his own query for his papers

- relational algebra

- theory-based query language that is really useful for explaining what real query languages do